

JUSTIN VARGHESE JOHN

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SUMMARY

Robotics and AI graduate researcher specializing in **machine learning, robot planning, learning-based autonomy and large-scale simulation**. Experienced in developing **symbolic and learning-driven planning systems**, high-fidelity simulators, and real-time robotic pipelines that scale from simulation to real-world deployment. Conducting master's thesis research in the AAIR Lab at Arizona State University.

EDUCATION

Master of Science in Robotics and Autonomous Systems (AI)

Ira A. Fulton Schools of Engineering

Arizona State University, Tempe

GPA: 4.0/4.0

August 2024 - Expected May 2026

AZ, USA

Bachelor of Technology (Honors) in Robotics and Automation

Minor in Computer Science and Engineering (AI)

APJ Abdul Kalam Technological University

GPA: 3.72/4.0

August 2019 - August 2023

Kerala, India

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Services Assistant (GSA) & Graduate Researcher

July 2025 - Present

Autonomous Agents and Intelligent Robots (AAIR) Lab, Arizona State University

Tempe, AZ

- Conducting master's thesis research under Prof. Siddharth Srivastava on learning **symbolic and relational abstractions for long-horizon robot planning** from raw, unannotated demonstrations.
- Developing **logic-based world models** and **high-level actions** and integrating learned symbolic representations with **PDDL-based task and motion planning** frameworks to enable **zero-shot generalization** to larger and more complex tasks.
- Designing and implementing **custom simulation and data collection domains** in **OpenRAVE** and **NVIDIA Isaac Lab**, supporting **scalable learning, evaluation and sim-to-real experimentation**.
- Re-architected** an existing **learning and evaluation pipeline** for **scalable parallel execution**, increasing **data throughput** and reducing **end-to-end training and evaluation time by ~90%**.
- Manuscript under submission to the Conference on Robot Learning (CoRL 2025).

Robotics Engineering Intern - EPICS Program

May 2025 - Aug 2025

Cartken

Tempe, AZ

- Performed **lifecycle and sustainability analysis** of autonomous delivery robots, contributing design recommendations that reduced projected material waste and improved component reuse potential.
- Assisted in **hardware debugging** and **prototype refinement, accelerating testing cycles** and **improving system stability** across development iterations.

SELECTED RESEARCH & PROJECTS

- ReflexDodgeBot: Real-Time Embedded Perception & Reactive Obstacle Avoidance**
Designed and deployed a vision-based robotic system capable of detecting and dodging fast-approaching objects in real time, achieving **30 FPS end-to-end** on NVIDIA Jetson Nano using monocular depth estimation, Kalman-based motion tracking and learning-based scene understanding.
- Autonomous Surveillance Robot System**
Developed an autonomous surveillance robot with vision-based perception, road-following, and obstacle avoidance, integrating closed-loop control and real-time decision making on embedded hardware, achieving **stable autonomous operation at 30 FPS** during continuous navigation.
- Autonomous Drone Control & Manipulation Digital Twins (MATLAB/Simulink)**
Built a high-fidelity **digital twin** of the MyCobot Pro 600 by **implementing forward and inverse kinematics, trajectory generation, and motion planning with closed-loop control**. Validated the full stack on **physical hardware** through autonomous maze-solving manipulation and designed and implemented vision-based **flight control algorithms** tested on a Parrot Mambo drone.
- Multi-Robot Exploration and Cooperative Mapping**
Implemented multi-robot exploration and mapping frameworks using **stochastic exploration strategies combined with potential field-based navigation**, achieving **over 90% environment coverage** with stable collision avoidance in simulation.
- Edge-Deployed Perception for Intelligent Transportation Systems**
Trained and optimized deep learning models for perception tasks such as **incident detection and semantic segmentation**, achieving **95% precision** and **75% model size reduction** for real-time edge deployment.

KEY SKILLS

Technical Skills

- Programming:** Python, C, C++, MATLAB, SQL
- Robotics & Planning:** ROS, OpenRAVE, Isaac Lab, SLAM, Task & Motion Planning, Autonomous Systems, 3D Perception
- AI / Machine Learning:** PyTorch, TensorFlow, Machine Learning, Computer Vision, Reinforcement Learning
- Systems & Performance:** Linux, CUDA, TensorRT, Parallel Computing, Embedded & Real-Time Systems
- Automation:** PLC Programming, SCADA, PID Control, Robotic Hardware & Actuation

Certifications

- Fundamentals of Engineering Project Management from Arizona State University
- Inter-Personal Skills in Engineering Project Management from Arizona State University